

SSH Brute-Force Investigation Report

Date: 21 May 2025 | Analyst: Sunil Rajpal | Project: Google Cybersecurity Certification

Executive Summary

An SSH brute-force attack was detected against an Ubuntu server. The attacker (**203.0.113.45**) made **150+ failed login attempts** targeting the root account over ~2 hours. **No successful breach occurred.**

Key Findings

Finding	Details
Attacker IP	203.0.113.45 (150 failures)
Normal baseline	2-5 failures per legitimate user
Primary target	root account (148 of 150 attempts)
Attack pattern	Concentrated burst — automated tool

MITRE ATT&CK Mapping

Technique	ID
Brute Force	T1110
Network Service Scanning	T1046

Recommendations

Immediate (Today)

- Block 203.0.113.45 at firewall
- Verify no successful logins from this IP

Short-term (This Week)

- Deploy fail2ban for auto-blocking
- Disable root SSH login (PermitRootLogin no)

Long-term (This Month)

- Enforce key-based authentication only
- Implement MFA for SSH access

Splunk Alert (Set up now)

```
bash

index=simulated "Failed password"

| bucket _time span=5m

| stats count by src_ip, _time

| where count > 10
```

IOC Summary

Type	Value
Malicious IP	203.0.113.45
Affected Service	SSH (TCP/22)
Targeted Account	root

Status: Contained | Risk: Low (attack failed) | Remediation: Ready